

CUSTOMER CHURN ANALYSIS REPORT

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Project Title: Customer Churn Analysis Dashboard

1. Introduction

Customer churn refers to customers who stop using a company's products or services. Analyzing churn helps organizations understand customer behavior and identify factors responsible for customer loss. This project presents an interactive Power BI dashboard that analyzes customer churn using the Telco Customer Churn dataset.

2. Objective

The objectives of this project are:

- Analyze customer churn patterns.
- Calculate churn and retention rates.
- Identify high-risk customer groups.
- Analyze payment methods and contract types.
- Compare internet service types with churn.
- Build an interactive Power BI dashboard.

3. Dataset Description

The dataset used for this project is the **Telco Customer Churn Dataset**.

Dataset Features

- Customer ID
- Gender
- Senior Citizen

- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn

Total records in the dataset: **7043**

4. Tools Used

Tool	Purpose
Power BI Desktop	Dashboard Development
DAX	KPI Calculations
Microsoft Excel	Dataset Source
Microsoft Word	Documentation

5. Exploratory Data Analysis (EDA)

The following analyses were performed:

- Customer Churn Distribution
- Gender Distribution
- Churn by Internet Service
- Customers by Contract Type
- Total Charges by Contract Type

- Customers by Payment Method
- Monthly Charges by Churn
- Customers by Tenure Group

6. Dashboard Development

The dashboard was designed using Power BI with interactive visuals and slicers.

The dashboard contains:

- KPI Cards
- Donut Chart
- Pie Chart
- Clustered Column Chart
- Bar Chart
- Horizontal Bar Chart
- Slicer
- Insights Section

7. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Total Customers
- Churn Customers
- Churn Rate
- Retention Rate



8. Key Insights

The analysis revealed the following insights:

- Approximately **26.58%** of customers have churned.
- The retention rate is approximately **73.42%**.
- Customers with **Month-to-Month contracts** have the highest churn rate.
- **Fiber Optic** internet users show higher churn compared to DSL users.
- Customers paying through **Electronic Check** have the highest churn.
- Customers with shorter tenure are more likely to churn.
- Long-term contract customers have better retention.

9. Conclusion

The Customer Churn Analysis Dashboard successfully identifies important factors affecting customer churn. The dashboard enables businesses to monitor customer behavior, understand churn patterns, and make informed decisions to improve customer retention.